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Here There Be Data: Mapping The Landscape Of Science

ScienceDaily (Apr. 7, 2004) — ARLINGTON, Va. -- In ancient maps of the world, expanses of unknown territory might hold a warning to would-be explorers: Here there be monsters. For today's explorers seeking to navigate and understand the world of science, the monsters are the untamed collections of data that inhabit a largely uncharted landscape.

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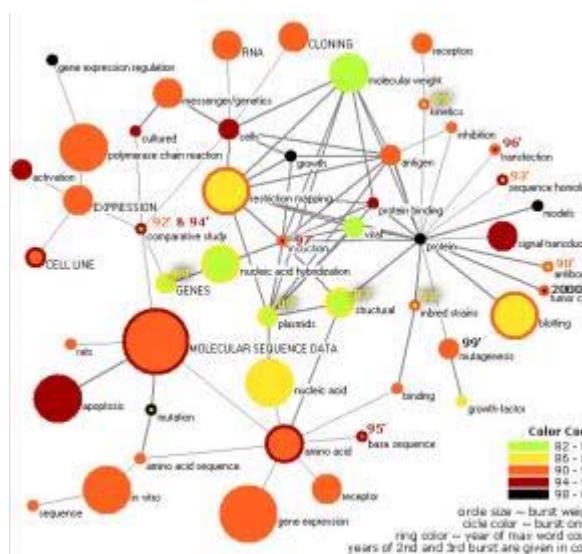
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The April 6, 2004, issue of the *Proceedings of the National Academy of Sciences* (PNAS) features nearly 20 articles by some of tomorrow's mapmakers. Representing the computer, information and cognitive sciences, mathematics, geography, psychology and other fields, these researchers present attempts to create maps of science from the ever-growing and constantly evolving ocean of digital data.

"Science is specializing at high speed, which leads to increasing fragmentation and reinvention," said Katy Börner of Indiana University. "Maps of publication databases or other data sources can help show how scientists and scientific results are interconnected."

College students might use such maps to see how well a syllabus covers a field's major topics, while companies could map out plans for targeting their investments. Funding agencies could keep an eye on research frontiers or forecast how funding decisions might affect a discipline. An online version could provide an effective interface to major databases.

"Ultimately, I'd like to see a map of science in schools, as common as the political world map," Börner said. "Continents' would represent the diverse areas of science, and closely related areas would reside on the same continent. Teachers



A map of the top 50 "hot" words in the most highly cited PNAS articles from 1982-2001. Words appearing more often have larger circles, while the circle color and ring color identify when the word first appeared and when its popularity peaked, respectively. (Credit: Ketan Mane and Katy Börner Indiana University.)

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Prepare to be Shocked

Knowledge Domains, sponsored by the National Academy of Sciences. Organized by Richard Shiffrin and Börner of Indiana University, the colloquium addressed the task of extracting meaningful and relevant information from largely unorganized data collections.

"Today, almost all of us access knowledge in ways vastly different from those used for hundreds of years," Shiffrin said. "The traditional method involved books, reference works and physical materials on library shelves, most of which had been verified for accuracy by one or another authority. Now, we sit at computers and cast our net into a sea of information, much of which is inaccurate or misleading."

Authors of 12 of the articles are supported by awards from the National Science Foundation (NSF), the independent federal agency that supports fundamental research and education across all fields of science and engineering.

Several of the papers describe ways to analyze article collections and map out landscapes that humans can view. Some methods, such as that proposed by Simon Dennis, "read" scientific articles and use a deep understanding of the content as the basis for a map. Other methods use relationship networks between the articles, such as citation of other papers, as the basis for a map.

"Process" models aim to better understand how the structure of scientific networks evolves over time. Filippo Menczer demonstrates that some combination of content and Web links or citation relationships needs to be considered, while Börner, Jeegar Maru and Robert Goldstone consider topics, newness, and linking to show how several such networks might evolve together.

Scientific landscapes might have hundreds of possible dimensions, presenting a challenge in creating two- or three-dimensional maps, according to Thomas Landauer and colleagues. Elena Erosheva and colleagues show that computer-derived mappings may not correspond to human-assigned categories and that more articles can be considered interdisciplinary than officially indicated by PNAS dual classifications.

Mapping methods must also identify the data-collection analogs of landmarks and borders. For example, Thomas Griffiths and Mark Steyvers found the "hot topics" that cropped up in a 10-year collection of PNAS articles. Similarly, Jonathan Aizen and colleagues describe how spikes in an item's Web popularity might be useful as time-sensitive landmarks.

The borders on these maps mark divisions between related scientific topical areas, groups of collaborators or other clusters that emerge from the data at hand. For example, Paul Cieniewicz

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Student-friendly GIS Leads To Real-world Science Inquiry

(Feb. 14, 2006) — A report by the National Research Council urging educators to teach K-1

students to think spatially and use geographic information systems (GIS) to do so underscores th

scientist may or may not be six degrees from Kevin Bacon, but Newman showed that scientists were about six coauthors away from any other scientist.

However, these borders, like the world's political boundaries, change over time. John Hopcroft and colleagues devised a method that mapped, across a landscape of 1.8 million computer science articles, the scientific communities that evolved over the course of a decade.

Finally, in a digital landscape with hundreds of possible options for north or south, east or west, drawing a map with which human explorers can navigate from point A to point B presents another set of challenges. Ketan Mane and Börner describe techniques to draw maps that highlight landmarks such as major research topics or trends. Alan MacEachren and colleagues show how techniques from geographic mapmaking might be applied to science landscapes.

"Creating a map for all of science will require large-scale cyberinfrastructure," Börner said. "The endeavor will involve terabytes of data—publications, patents, grants and other databases—scalable software and large amounts of number-crunching power. Such computational effort is common in physics or biology but not in the social sciences. However, maps of science will benefit every field."

The research in the following papers in the April 6 issue of PNAS was supported in whole or in part by awards from the National Science Foundation:

- * Jonathan Aizen, Daniel Huttenlocher, Jon Kleinberg, and Antal Novak. "Traffic-based feedback on the web."
- * Katy Börner, Jeegar T. Maru, and Robert L. Goldstone. "The simultaneous evolution of author and paper networks."
- * Simon Dennis. "An unsupervised method for the extraction of propositional information from text."
- * Elena Erosheva, Stephen Fienberg, and John Lafferty. "Mixed-membership models of scientific publications."
- * Paul Ginsparg, Paul Houle, Thorsten Joachims, and Jae-Hoon Sul. "Mapping subsets of scholarly information."
- * Thomas L. Griffiths and Mark Steyvers. "Finding scientific topics." John Hopcroft, Omar Khan, Brian Kulis, and Bart Selman. "Tracking evolving communities in large linked networks."
- * Thomas K. Landauer, Darrel Laham, and Marcia Derr. "From paragraph to graph: Latent Semantic Analysis for information visualization."
- * Ketan K. Mane and Katv Börner. "Mapping topics and topic

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